

[Oct 22/TE/Insem]-526

T.E. (Computer Engineering)

THEORY OF COMPUTATION

(2019 Pattern) (Semester -I) (310242)

Time : 1 Hour]

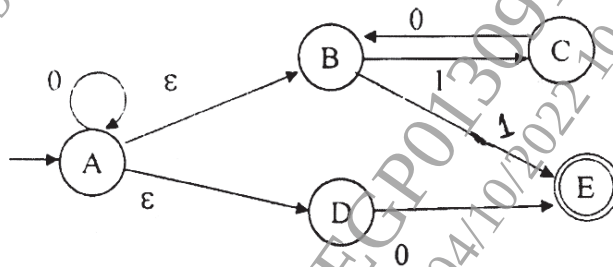
[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Convert the given NFA- ϵ to an NFA to DFA.

[10]



b) Define Pumping Lemma and apply it to prove the following

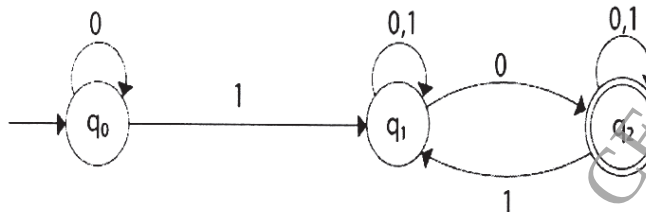
 $L = \{0^m 1^n 0^{m+n} \mid m \geq 1 \text{ and } n \geq 1\}$ is not regular

[5]

OR

Q2) a) Convert following NFA to DFA

[6]



b) Design a Mealy machine that accepts strings ending in '00' or '11'.
Convert the Mealy machine to the equivalent Moore machine

[9]

P.T.O.

Q3) a) Convert the following RE to ϵ -NFA and find the ϵ -closure of all the states and corresponding DFA. $(0+1)^*. 1.(0+1)$ [9]

- b)
- The set of strings over $\{0,1\}$ that have at least one 1. [6]
 - The set of strings over $\{0,1\}$ that have at most one 1.
 - The set of all strings over $\{0,1\}$ ending with 00 and beginning with 1.

OR

Q4) a) Consider the two RE $r=0^*+1^*$, $s=01^*+10^*+1^*0+(0^*1)^*$ [8]

- i) Find the string corresponding to r but not to s .
ii) Find the string corresponding to s but not to r .
iii) Find the string corresponding to both r & s .
iv) Find the string corresponding to neither r nor s .
- b) Write regular expressions for the following languages over the alphabet $\Sigma=\{a,b\}$ [7]
- i) All strings that do not end with 'aa'.
- ii) The set of all strings ending neither in b nor in ba
- iii) Find the shortest string that is not in the language represented by the regular expression $a^*(ab)^*b^*$.



Total No. of Questions : 4]

SEAT No. :

P-5341

[Total No. of Pages : 3

[6187]-426A

T.E. (Computer Engineering) (Insem.)
THEORY OF COMPUTATION (Theory)
(2019 Pattern) (Semester - I) (310242)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer the question of 1 or 2, 3 or 4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Draw FA for the following language over $\{0, 1\}$ **[8]**

- i) Number of 1's is multiple of 3.
- ii) Number of 1's is not multiple of 3

b) Covert following NFA into equivalent DFA and perform DFA minimization **[7]**

Q/ Σ	0	1
$\rightarrow P$	{P, Q}	{P}
Q	{R}	{R}
R	{S}	--
S^*	{S}	{S}

OR

Q2) a) Construct DFA for checking "whether a string over alphabet $\{a, b\}$ contains a substring aba ". **[5]**

- b) i) Differentiate between Moore machine and Mealy machine.
- ii) Construct Moore machine equivalent to the following Mealy machine.
(Show it in transition Diagram)

$M = (Q, \Sigma, \Delta, \delta, q_0)$ where $Q = \{q_0, p_0, p_1\}$, $\Sigma = \{0, 1\}$, $\Delta = \{y, n\}$ and δ is shown as given below.

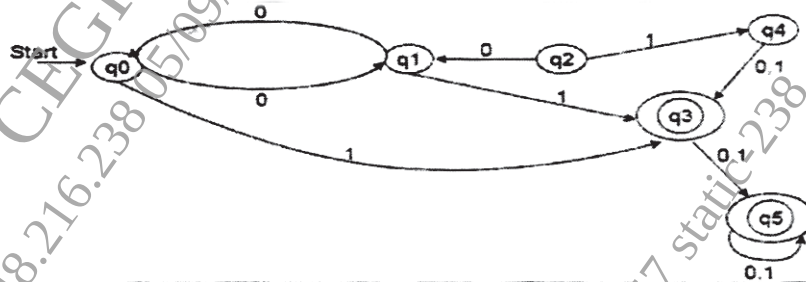
P.T.O.

States	Input / Output	
	0	1
q0	p0/n	p1/n
P0	p0/y	p1/n
P1	p0/n	p1/y

[5]

c) Convert the following DFA to its Minimized form (Minimization of DFA).

[5]



Q3) a) Prove that LHS RE is equivalent to RHS RE

$$(1+00^*1)+(1+00^*1)(0+10^*1)^*(0+10^*1)=0^*1(0+10^*1)^*$$

[5]

b) Find a regular expression corresponding to each of the following subsets of $\{0,1\}^*$

[6]

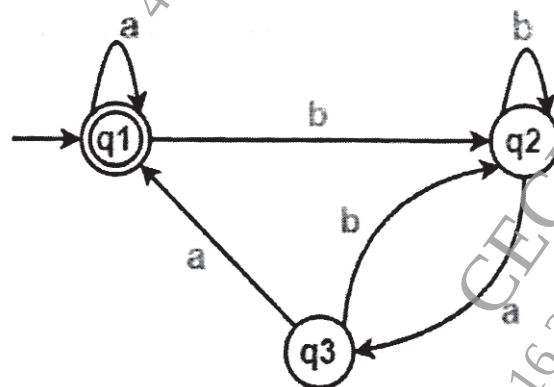
- The language of all strings containing exactly two zeros
- The language of all strings containing at least two zeros
- The language of all strings that do not end with 01.

c) Write a note on Myhill Nerode theorem.

[4]

OR

Q4) a) Construct Regular expression for following DFA using Ardens theorem.[7]



- b) i) Write regular expression for a set of strings of 0s and 1s with even number of 0s.
- ii) Write regular expression for a set of strings of 0s and 1s containing odd number of 1s.

[4]

- c) Choose any one option given below and give the justification “The regular expression $0^*(10^*)^*$ denotes the same set as” [4]

- i) $(1^*0)^*1^*$
- ii) $0^+(0+10)^*$
- iii) $(0+1)^*10(0+1)^*$
- iv) none of these

▽▽▽▽

T.E. (Computer Engineering) (Insem)
THEORY OF COMPUTATIONS
(2019 Pattern) (Semester- I) (310242)

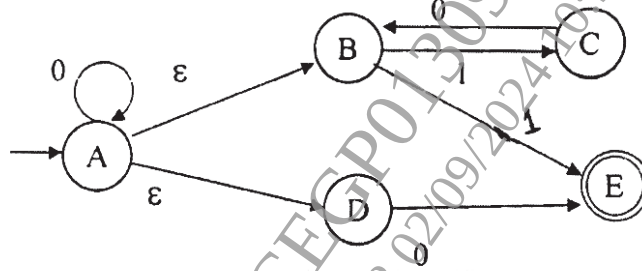
Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer the question of 1 or 2, 3 or 4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

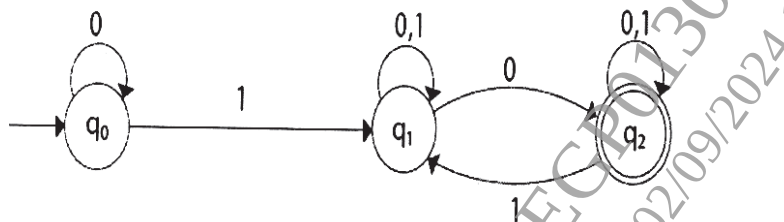
Q1) a) Convert the given NFA- ϵ to an NFA to DFA. **[9]**



b) Design a DFA which can accept a decimal number divisible by 3. **[6]**

OR

Q2) a) Convert following NFA to DFA. **[7]**



- b) i) Design a moore machine for the 1's complement of binary number.
- ii) Design a Mealy machine to find out 2's complement of a given binary number. **[8]**

P.T.O.

Q3) a) Convert the following RE to ϵ -NFA and find the ϵ -closure of all the states and corresponding DFA. $(0 + 1)^* \cdot 1 \cdot (0 + 1)$. [9]

- b) i) Regular Expression of strings over $\{0,1\}$ that have at least one 1.
 ii) Regular Expression of strings over $\{0,1\}$ that have at most one 1.
 iii) Regular Expression of all strings over $\{0,1\}$ ending with 00 and beginning with 1.

[6]

OR

Q4) a) i) Write the regular expression for the language starting with a but not having consecutive b's.

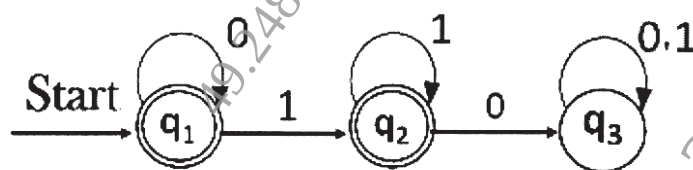
ii) Write the regular expression for the language L over $\Sigma = \{0,1\}$ such that all the string do not contain the substring 01.

iii) Write the regular expression for the language containing the string over $\{0,1\}$ in which there are atleast two occurrences of 1's between any two occurrences of 1's between any two occurrences of 0's.

[7]

b) Design a FA from given regular expression $10 + (0 + 11) 0^* 1$. [4]

c) Construct the regular expression for the given DFA using Ardens Theorem. [4]



Total No. of Questions : 4]

SEAT No. :

PE326

[Total No. of Pages : 2

[6579]-326

T.E. (Computer Engineering)/(Computer Science) (Insem)

THEORY OF COMPUTATION

(2019 Pattern) (Semester - I) (310242)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

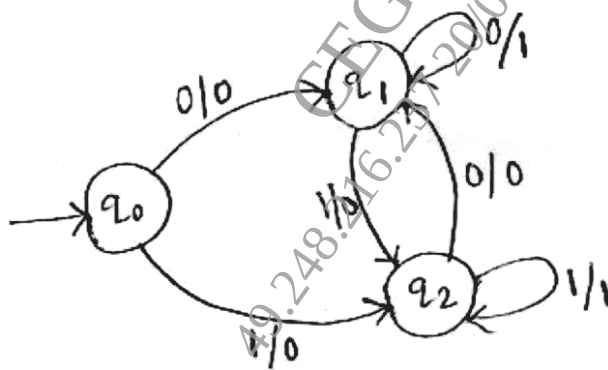
- 1) Attempt Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data if necessary.

Q1) a) Design DFA which accepts set of strings over alphabet $\Sigma\{a, b\}$ such that [8]

- i) if it contains exactly 3 number of a' s.
- ii) if it contains at least 3 number of a' s.

b) Consider the following Mealy Machine of Construct equivalent Moore machine. [7]

Also differentiate between Moore and Mealy machine (any 4 points).



OR

Q2) a) Design Moore machine such that for every substring that ends in bab the machine will give output 1 over alphabet $\Sigma\{0, 1\}$. Further convert the same Moore machine into Mealy machine. [8]

b) Design a DFA which can accept a binary number divisible by 3. Explain the logic also. [7]

P.T.O.

Q3) a) Write a regular expression to accept following language over alphabet $\{a,b\}^*$ [6]

- i) Strings having at least one occurrence of substring 'aaa'.
- ii) Strings starting and ending with same symbol.
- iii) Strings having even number of a's.

b) Using Arden's theorem, find regular expression. [5]



c) Draw NFA with epsilon Moves for RE = (a^*b^*) [4]

OR

Q4) a) Check the equivalence of the Regular Expression. [6]

- i) $(a^*bbb)^*a^*$ & $a^*(bbba^*)^*$
- ii) $((a+bb)^*aa)^*$ & $E+(a+bb)^*aa$

b) Describe the languages accepted by the following regular expression and justify. [5]

- i) $a(a+b)^*ab$
- ii) $(1^*01^*01^*)^*$

c) Show that $L = \{a^n | n \text{ is a prime}\}$ is not regular. [4]

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